



INTERNSHIP REPORT

Kohinoor Textile Mills Private Limited

26-Jun-2024 to 07-Aug-2024

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Approval Letter

This is to certify that Muhammad Huzaifa Internship Project Report titled “Develop Baseline of Oracle Discrete Manufacturing (ODM) for Stitching Department: Reviewed each process of stitching department from cutting received to transferring packed goods to warehouse”.

Muhammad Huzaifa from Industrial Engineering Department, registration number 21-IE-27 collaborated with the Made-ups department to prepare the internship report. I solemnly clarify that the data in this internship report is very genuine. I strongly suggest their report for higher academic recognition. Student is morally solid and have pleasant attitude. It has a pleasure to work with him. I wish him the success in his future.

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Abstract

In KTML I have done my 6 weeks of Internship. Here at KTML in first week I learned the overall process flow of textile industry from my supervisor. In 2nd and 3rd week I start learning about the working process of the Made-ups section for which I spent time in folding and cutting section, their I learned about Machine, Documentation, staff hierarchy and work process. In addition, I spent most of my time in stitching section and learned the things as I did in cutting. Afterwards in my 4th week of internship I have spent my time in embroidery, quilting and multi-needle embroidery. In 5th and 6th week, I started working on my project for which I have again gone through the documentation process flow-taking placing in the Unit 1 of stitching section. I tried to implement the Paper free documentation procedure so that the documents can be shifted in the soft form. More how I have also performed the SAM study to analyze the time of different operations and categorize them accordingly.

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Chapter 1: About Textile and Kohinoor

1.1 Definition

Textile is concerned with the production of fabric and cloth for the daily use. There are different types of textile industries:

1. Home Textile
2. Hosiery
3. Apparel
4. Sports Textile
5. Geo Textiles

1.1.1 Home Textile

Home Textile mainly deals with the home fabrics like Bedding (sheet, blankets, and comforters), Towel, Curtains, Upholstery fabric, Rugs, Mattress ticking.

1.1.2 Apparel Textile

It mainly includes the wearing type of fabrics like casual wears, formal wears, Active Wear, Lingerie, outerwear.

1.1.3 Sports Textile

It mainly includes the sports equipment in which fabric is used like shoes, protective gear, and active gear.

1.1.4 Geo Textile

It mainly includes the fabrics involved in erosion control, road construction, landscaping.

1.2 Textile Industries in Pakistan

1. Interloop
2. Nishat Mills Limited
3. Sapphire Textile Mills
4. Gul Ahmed Textile
5. Style Textiles
6. Kohinoor

1.3 Top Exporters in World

1. Vietnam
2. India
3. China
4. Bangladesh

1.4 Main Process in Textile

1.4.1 Spinning

It is the process, which involves the manufacturing of yarn from raw material like synthetic equipment or non-synthetic equipment, which is normally cotton or polyester.

1.4.2 Weaving

It involves the production of fabric from yarn, which is known as Geirge fabric. Geirge Fabric are normally of two types:

1. Knitted Fabric
2. Weaven Fabric

1.4.3 Dying and Printing

It involves dying Geirge fabric because geirge fabric is grey and it is dyed and printed to create a proper colorful fabric.

1.4.4 Cutting

It involves cutting a large fabric into small cut fabric component according to the cut size of fabric component.

1.4.5 Stitching

Then cut fabric is stitched according to the requirement of the garment or requirement of customer.

1.4.6 Checking

In this final stitched fabric is checked properly that all the stitch size is up to the mark and fabric is clearly checked and the needle holes on the fabric is checked on the fabric.

1.4.7 Packing

It involves wrapping, compressing of the stitched and checked fabric.

1.5 About Kohinoor

In Kohinoor, there are main work of Home Textile. Kohinoor was established in 1953 and it is one of the oldest companies in Pakistan. It is vertically integrated industry, which has become a key player in countries textile industry.

The KTML is the part of renowned Maple Leaf Group, which has diversified interest in different sectors. Its manufacturing facility is in Rawalpindi, Pakistan and covers a large area of of weaving, processing and finishing of various types of textile.

The Kohinoor is involved in the production of home textile, yarn and fabric. Over the years, Kohinoor have earned a profound reputation in the manufacturing of textile equipment both in domestic and international market.

KTML has remained committed in using strong equipment of machineries, tools and skilled professional, which has helped Kohinoor to build a strong reputation in the eyes of the customer because of their outstanding quality.



Figure 1.1: Kohinoor Textile Mills

1.6 Kohinoor Customers

Kohinoor customer are worldwide which mainly are Laura Ashley, Indigo, Lacoste e.t.c. Also there is the Fair Price Shop where the Home Textiles are sold. Which means the Kohinoor both sells in international market and local market.



Figure 1.2: Fair Price Shop

Chapter 2: Folding

2.1 Definition

Folding is the first and foremost process before cutting as in this the processed fabric is examined through folding machines. The examination is done in the variety of ways that includes color, dye, shades and quality of the fabric. Through folding the length of the fabric is also measured to be rolled in further paper rolls according to the requirement.

2.2 Staff Hierarchy

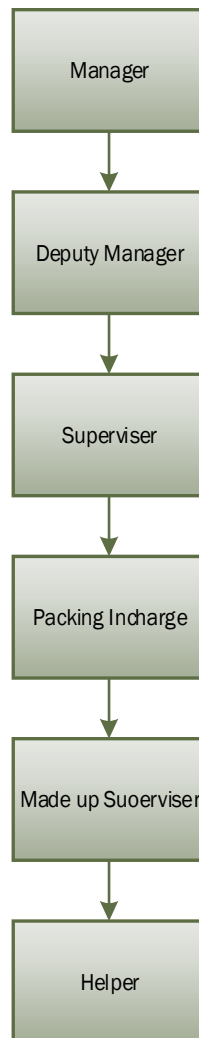


Figure 2.1: Staff Hierarchy of Folding

2.3 Explanation

In folding the main process is to check the upcoming fabric from processing section or the most recent shrinkage machine through which tension in the fabric is controlled. Then the open fabric is set on the rolling inspection machine and the process of folding is done on it.



Figure 2.2: Rolling Inspection machine

2.4 Details of Folding Machine

The name of the machine used for folding is **rolling inspection machine**.

There are **9 rolling inspection** machine in the folding section.

There is 1 operator on each machine.

2.5 Things Checked in Folding

Following are the standard checked during folding:

1. Customer Standards
2. Comparison is done
3. Hand Feel
4. Shade Standard
5. Design Standard

2.6 Some important notes

Roll of thick roll is 150m.

Roll of thin roll is 300m.

However, in both of the cases the weight of roll is equal.

There is 50,000 m capacity per day.

2.7 Jasipani

It is a document that develops a link between production unit and customer and on this document all the details regarding fabric are mentioned.

2.8 Continuity Card

A continuity card is a card of shade identification in which the shades identified through shade checkers and on that continuity card the fabric of different shades attached identifying about the shade.

Normally two to three shades identified which attached on the continuity card.

The shade checking procedure is quite simple in which there is a one room specified where under different lights shades identified. Those lights vary on the type of articles being checked.

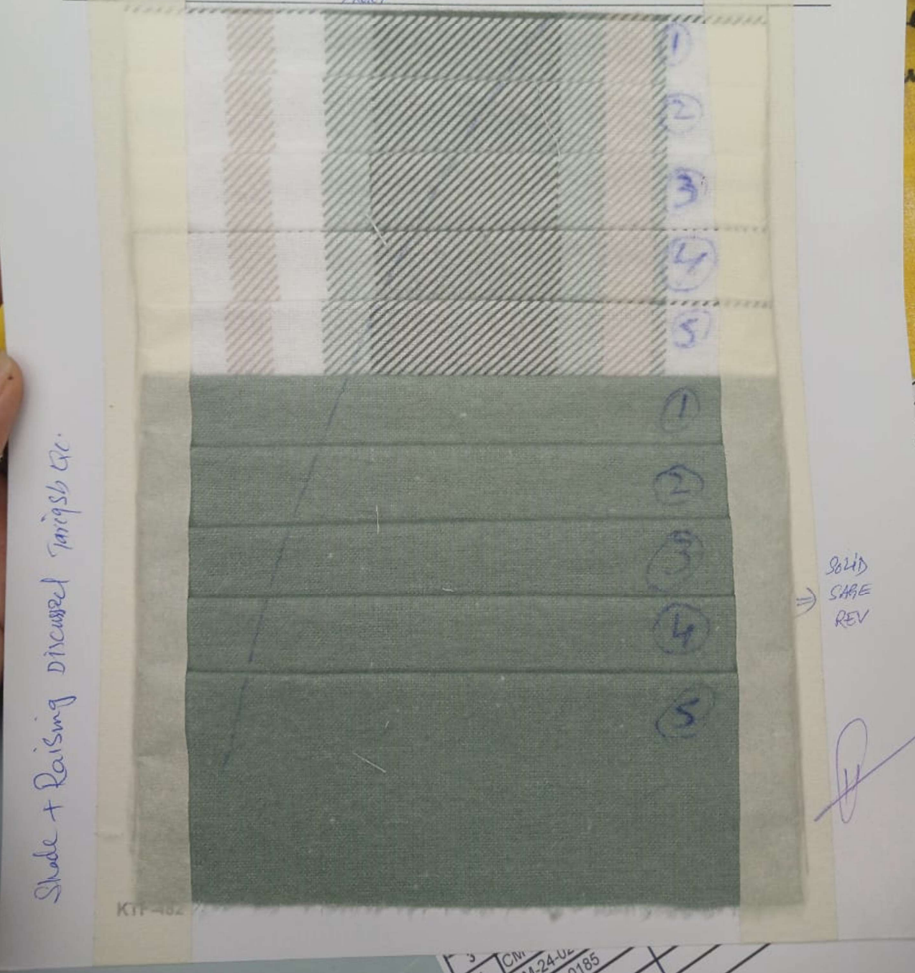
Those lights are UV, Day Light, Warm Light, Blue Light, indigo and these lights are assembled on the light checking table.

EFF

KOHINOOR TEXTILE MILLS LTD RAWALPINDI

CONTINUITY CARD

CONT. NO. <u>CM-272</u>	DATE <u>5-07-2014</u>
QUALITY <u>CT-2020/4442</u> <u>96</u>	CUSTOMER <u>Wadia Home</u>
DESIGN NO/SHADE <u>SWT</u> <u>PLAID</u> <u>SAGE</u>	CONSTRUCTION <u>CT-2020/4442</u>
BATCH NO. <u>(5)</u>	



Shade + Raising Discussed Taring Sb Etc.

Solid
SAGE
REV

KIT-432

CM-24/44
24-0185

Figure 2.3: Continuity Card

2.9 American 4.0 System

American 4.0 system is the type of inspection system in which the fabric on rolling inspection machine is inspected and is graded with respect to this American 4.0 System. This system mainly suggest the penalty points on the amount of the defect detected in inches. The detail are mentioned in the figure below:

KOHINOOR TEXTILE MILLS RAWALPINDI PAKISTAN

4-Point Inspection System

Size of Defect	Penalty Point
3" or Less	1
Over 3" but less than 6"	2
Over 6" but less than 9"	3
Over 9"	4

Note :

1 Meters = 1.0936 Yards = 3.2808 Feet = 39.37 Inches = 100 Centimeters.
 1 Yard = 0.9144 Meters = 3 Feet = 36 Inches = 91.44 Centimeters.

22 Penalty Point per 100 square Meters.
 No more than four penalty point may be assigned for any single defect in a meters.
 No more than four penalty point may be assigned to one linear meters.

Maximum Acceptable Points

Points per 100 square meters.

Calculations:

$$\text{Points}/100\text{M}^2 = \frac{\text{Total Points per Roll} \times 100}{\text{roll in meters} \times \text{Width of roll}}$$

OR

$$\text{Points}/100\text{M}^2 = \frac{\text{Total Points} \times 39}{\text{Total Mtrs in Roll} \times \text{Width of roll}}$$

COLOR BAND MAKING PROCEDURE

Collection of Continuity

Figure 2.4: American 4 Pont System

2.10 Roll Tag

After the rolls are made then the roll tag is attached on it on which all the details are mentioned on it regarding the roll, which includes size, origin country and much more.



Figure 2.5: Roll Tag

2.11 Sample Shade

In the variety of shades one best shade is selected that matches the customer requirement closely and is set as a standard shade and all the details are then mentioned on it regarding the CM.

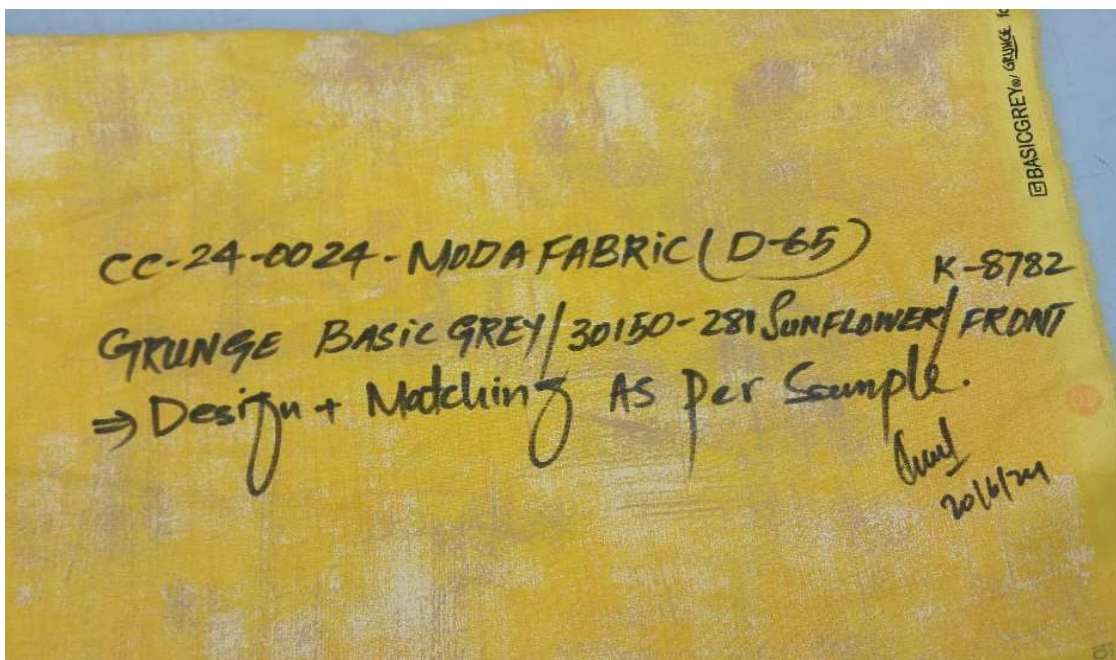


Figure 2.6: Sample Shade Fabric

2.12 Color Band Making Procedure

It goes in a step-by-step hierarchy as showed in the figure:

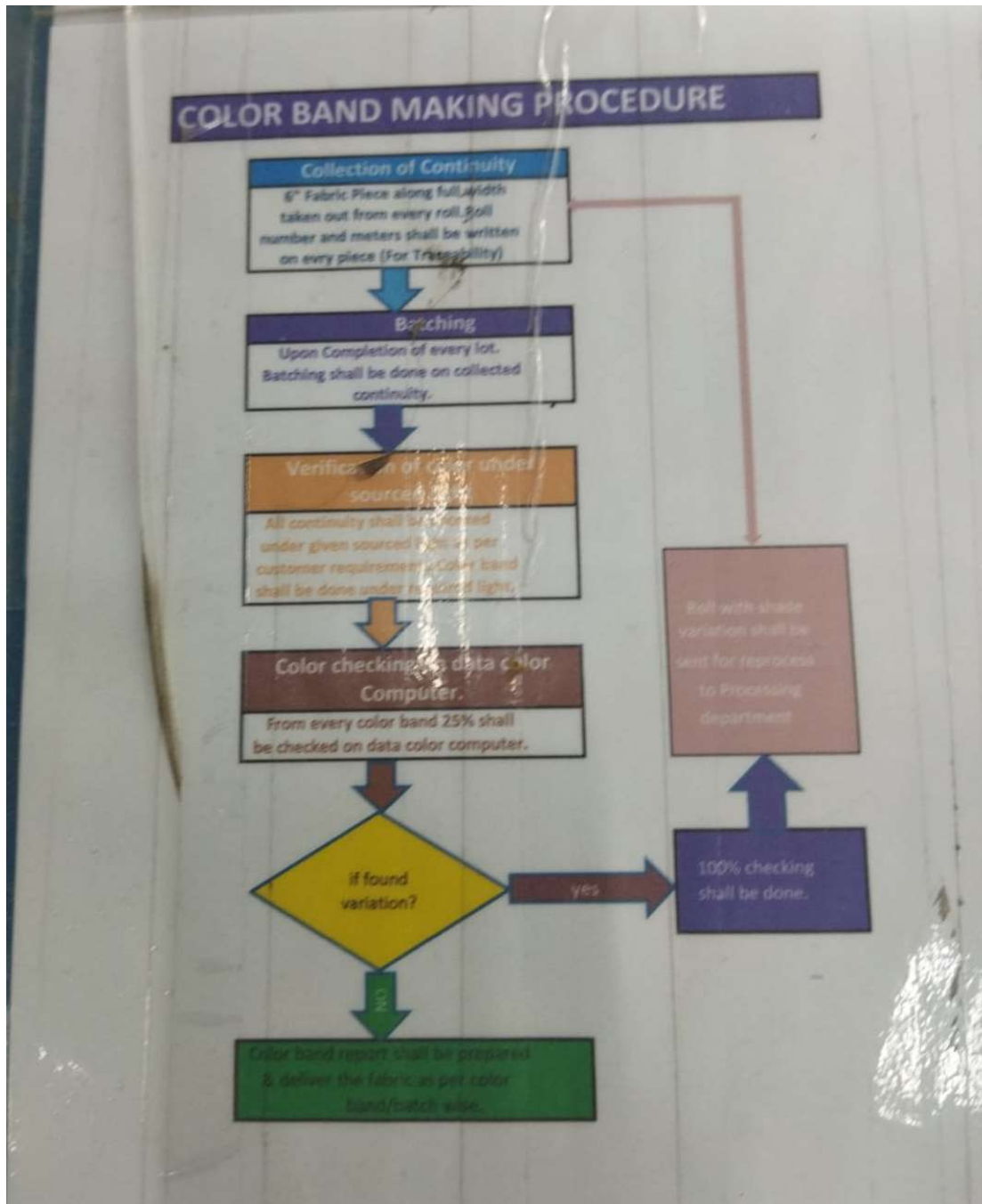


Figure 2.7: Color Band Making Procedure

2.13 Folding Defects

Following are some common folding Defects;

1. Stain
2. Misprint
3. Shade Difference
4. Registration Out

Chapter 3: Cutting

3.1 Defination

Cutting is the process of cutting fabrics into patterns for use in making home textile equipment. This process involves laying out patterns on fabrics and cutting along the lines to create the different parts of a home textile equipment.

Cutting is the main process of made ups in which the cloth coming from folding is converted into useful home textile equipment's using different cutting machines.

These cutting machines are helpful for making the cloth useful for stitching. The cutting process can be done manually or with the help of machines.

3.2 Staff Hierarchy

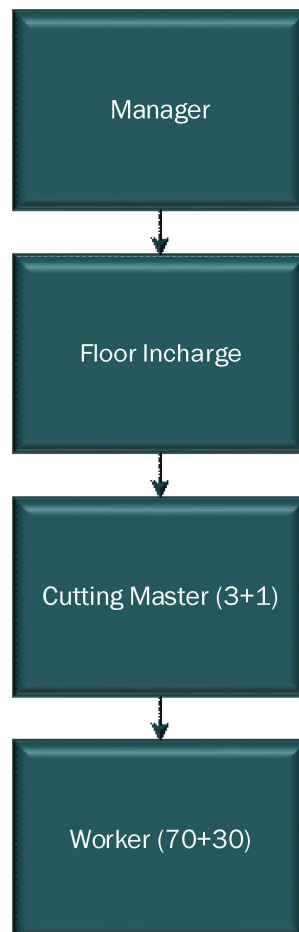


Figure 3.1: Staff Hierarchy

The staff hierarchy goes in the step-by-step manner:

3.2.1 Manager

There is only one Manager who manages all the working process in the cutting floor. Those process can be related to either worker, machine or material.

3.2.2 Floor In charge

He manages all the operations happening on the floor as instructed by the manager and convey it to the concerned on duty Master to manage the particular work.

3.2.3 Master:

There are three master in the morning shift and the two master in the night shift who supervise the workers with their concerned task and check all the work being done that either that work is done upto the mark according to the demand of customer. They also write all the processes and record on the cutting report.

3.2.4 Worker:

The worker works on their articles to cut and task allotted to them either those task involves monitoring or any operation of work allotted to them.

3.3 Process Flow:



Figure 3.2: Process Flow

In Cutting Section firstly the fabric is issued from the stacking area with the help of cutting plan.

Then the process of folding is done in which the fabric length is noted and all the minor details of the fabric is checked which mainly includes the quality of print and color shades and their the CM number plays a vital role which helps the fabric to be distinct with the other fabrics so that is one set of article the shade remains the same means if pillow have a shade then the bedsheet of that pillow must have the same shade.

Then the fabric is loaded on to the fabric for layering which is done with the help of layer man.

Then the first two layer inspection is done and then the marking is done which is checked and then the cutting is done and again the quality of cut is checked by quality representatives.

After that the marking is done for further article wise cutting and marking is checked and then again cutting is done.

Further more bundling is done and then counting by 3 TCB methodology and then according to Master Wise Report the cut bundles are sent to the stacking area for stitching process.

3.3.1 Machines used in Cutting Section

Table 1: Machines used in cutting

Machines	Quantity
Gerber Layering Machine	1
Houser Slitting Machine	1
Kansai Stitching Machine	4
Overlock Machine	4
Scalup Machine	1
Roll Cutting Machine	1
Slitting Machine	1 manual, 1 automatic

3.3.2 Cutting Process

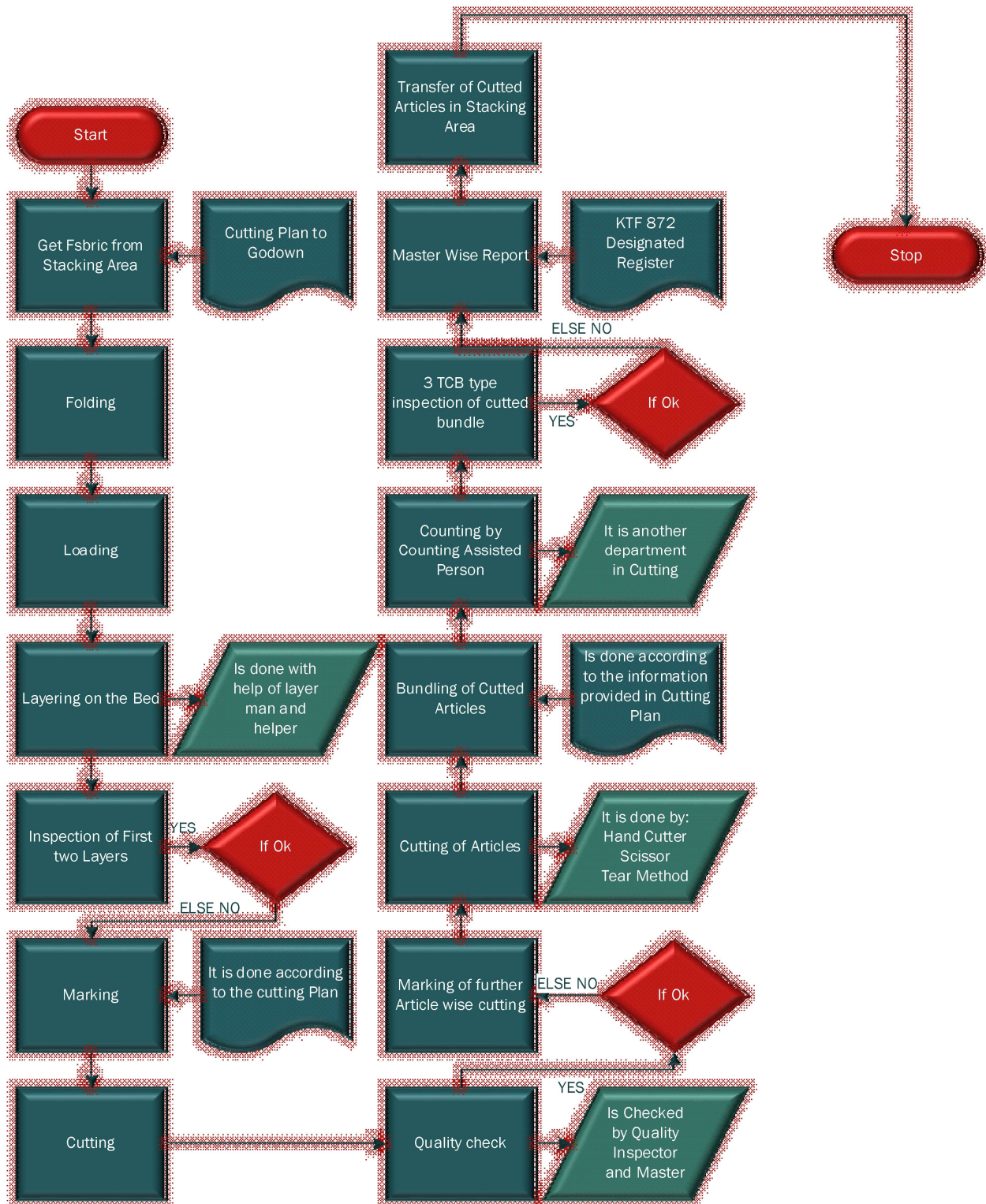


Figure 3.3: Cutting Process

3.4 Machines

Machines are categorized into two types used for cutting which are manual cutting machines and automated cutting machines:

3.4.1 Automated Cutting Machines

An automated cutting machine is a computer controlled device that uses sharp cutting tools to cut fabrics with precision and efficiency. These machines are programmed with specific instructions to control the cutting process, ensuring accurate and repeatable results.

The main purpose of automated cutting machine is to overcome or minimize the human effort and the automated cutting machines that were used includes:

- Gerber Layering Machine
- Houser Slitting Machine
- Slitting Machine
- Bias Roll Cutting

3.4.1.1 Gerber Layering Machine

Gerber layering machine are automated spreading systems that are used for technical textile production. They are designed to optimize material utilization by accurately aligning fabric edges. It layers the cloth from roller, is placed on the bed, and is cutted manually according to CM. It has one roller. It is a semi-automatic layering machine. It may have a capacity of layering 8000-meter fabric per hour. Its bed length is 98.4 feet and width is 11.48 feet.



Figure 3.4: Gerber Layering Machine

3.4.1.2 Houser Slitting Machine

In this the layering is done in the same way as of Gerber Layering Machine but the main difference is that its cutting is also automated.

It has two rollers.

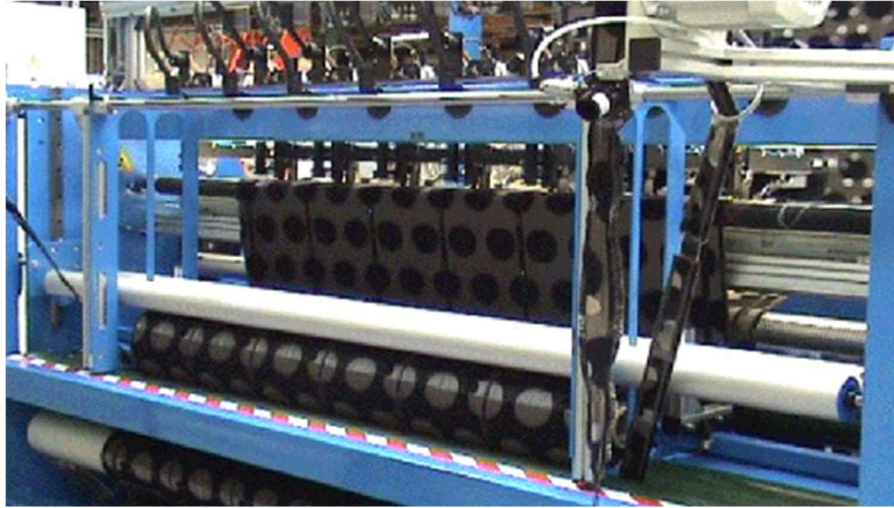


Figure 3.5: Houser Slitting Machine

3.4.1.3 Slitting Machine

A slitting machine also known as slitter is a specialized piece of equipment designed to wide rolls of material into narrower strip.

The main purpose of this machine is to make long strip in the form of Rolls according to the dimension provided by CM.

On Slitting Machine, rolls are loaded by rolling it with a tape on it so that while cutting the cut does not make fabric slaughter or damaged.

Slitting machine is both manual and automated well the main difference between the automate and manual slitting machine is the difference of the sharpening the blade automatically after each cut and the auto programming that is both available in automated slitting machine but not in manual slitting machine.

There is also one important point that the complex designed fabrics are firstly overlocked and then is loaded on to the rolls for slitting while the plain or simple designed fabrics are directly loaded on to the rolls for slitting without the overlock.



Figure 3.6: Slitting Machine

3.4.1.4 Bias Roll Cutting

This machine is used to make rolls for slitting purpose for this firstly the fabric is loaded on it and then it is also helpful for doing taping on fabric as it roll the roller and helps the worker to roll the tape on the fabric so that the slitting can be done properly. The rolling done according to the plan. Its purpose is to roll the fabric both angle wise and straight according to the requirement.



Figure 3.7: Bias Roll Cutting Machine

3.4.1.5 Scallop Cutter

Automatic cutting machine designs the scallop shaped design and embroidery on the fabric according to the dimension of the scallop provided to it. The machine works with the circular knives that adjust to the thickness of the fabric.



Figure 3.8: Scallop Cutting Machine

3.4.2 Manual Cutting Machine

Manual Cutting Machines are those machines that require human effort for their operation. They are chosen according to the type of fabric and requirement of the cut.

Example:

1. Scissor
2. Hand Cutter
3. Tear Method

3.4.2.1 Scissor

Scissors are a cutting tool that are used to cut fabrics and various thin material like paper but in textile sector scissors are helpful in cutting the small articles like pillows or small pieces of fabrics but through scissors it's only possible to cut minimum layers of fabric to have a high quality of cut and to maintain precision is a bit difficult as compared to other automated cutting tools.

3.4.2.2 Hand Cutter

A hand cutter is a semi-automated tool that is used for cutting the fabrics. It is used to cut a single layer of fabric in a straight line to create layers. Normally it is used to cut big articles to make layers of it like bedsheet and fitted.

3.4.2.3 Tear Method:

This method actually works for the thin fabric and is the easy method of cutting if the requirement of cut is in the straight line. A small cut is created with scissor and along the cut rest of the cut is created through the tear method in a straight line.



Figure 3.9: Scissor



Figure 3.10: Hand Cutter



Figure 3.11: Cutting Tools

3.4.3 Sewing Machines

3.4.3.1 Kansai Stitching Machine

The Kansai Stitching machine is used to create the stitch to create the design so that the designs can be made by stitching. Normally on the kansai-stitching machine, the frills of curtains are stitched so that the designed can be made on them to create luxurious look.

If we talk about the working details or accessories then it can be said that:

- 1 It can run on maximum of 12 needles and 24 threads.
- 2 It has also 12 bobbins.

3.4.3.2 Overlock Machine

The overlock machine is designed to create the overlock type stitch on the fabric for the strong stitch. It normally is done at the edges of the cloth for joining two fabrics with the strong bond. Sometimes it is also done before the slitting process for making the design in

a perfect straight hierarchy. If we talk about the accessories that are included in the overlook then they are:

- 1 5 Threads
- 2 4 Loopers

3.5 Process Document

Process document mainly includes the plan or schedule on which the cutting process must be done. The process document includes the following terms that is followed by the operator of the machine

- ➔ Color
- ➔ Article Size
- ➔ Article Shape
- ➔ Article Name

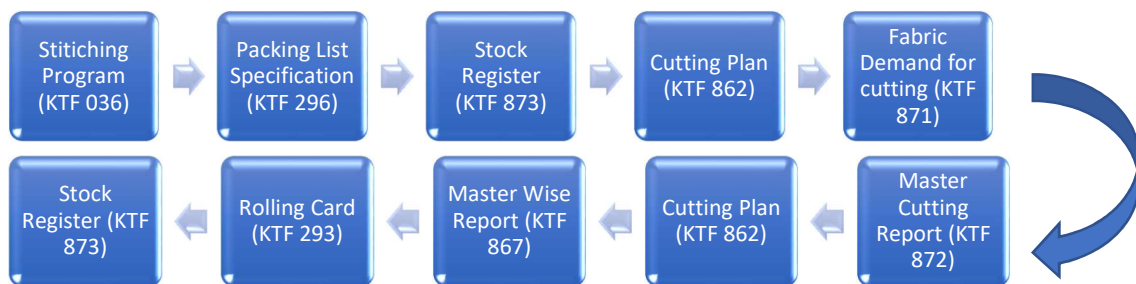
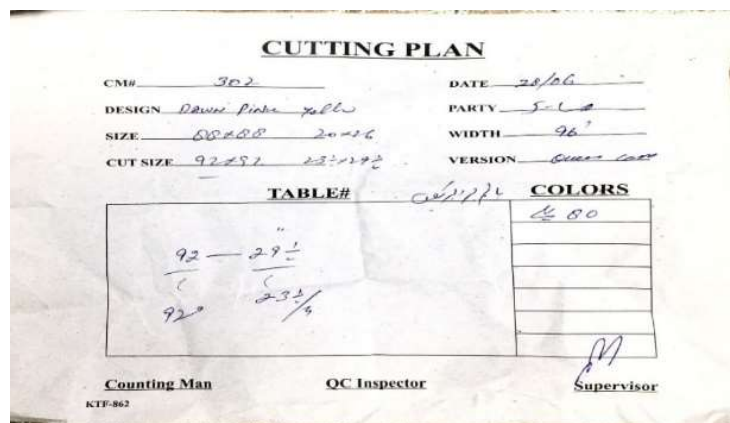


Figure 3.12: Process Document Hierarchy

3.5.1 Cutting Plan

Cutting Plan is allotted for every article with reference to its CM number for the clarification of all the details related to the article being cut such as CM number, design, date, size, cut size, party, width, version e.tc.



CUTTING PLAN

CM# 302 DATE 20/06

DESIGN Dawn Pink 486 PARTY 5-L

SIZE 80-88 20-26 WIDTH 96"

CUT SIZE 92-98 23-29 VERSION Quart 1st

TABLE# 92-29 1/4 COLORS 480

Counting Man KTF-862 QC Inspector [Signature] Supervisor [Signature]

Figure 3.13: Cutting Plan

3.5.2 Master Cutting

Master cutting Plan is noted on the relevant Master KTF numbered register, which includes all the related information regarding articles made at the time of the concerned Masters Shift. The information is listed below:

MASTER CUTTING						
M/S	ELIS		Program Date			
Design:	WHITE		Quality 40/40 110x80			
CM #:	258		PLAIN			
P/D/ No.			Week			
Article Name	Colour Version	Article Size CM	Cutting Size CM	Consumption Per Piece	Width	Required Piece
FLAT		305x315	305x325	7	120	2000
11		245x315	245x325	7	96	600
816		155x240	(159x246) ²	7	96	960
5		255x260	(259x266) ²	7	106	1500
2	58	290x260	(294x266) ²	7	106	1286
EXP-PL	FLAT 100%	54x81.420	65x204	7	106	12224
		55x100	114x105	7	96	4640
		57 1/2				
REMARKS						
Fabric Received	=	5	96	120		
Fabric Required	=	106	7611	6630		
Fabric Cutt	=					
Fabric RTG	=	7124	1-1909	5056		
		7029	2-1316	1585		
		3034	3-1146			
		4843				
		4371				
		2203				

KTF-872

Figure 3.14: Master Cutting Plan

3.5.3 Continuity Card

It is used to check the relevant shade of the article being made through continuity card which helps in comparing the shade of the fabric of different rolls that either the articles of a set either belong to the same shade or not so this continuity card is fruitful in understanding the shades.

Q/C

KOHINOOR TEXTILE MILLS LTD RAWALPINDI
CONTINUITY CARD

DATE 27-06-24

CONT. NO. CM-302 CUSTOMER JLA HOME

QUALITY CT 3030/7660 96" CONSTRUCTION CT 30x30/76x60

DESIGN HO/SHADE DAWN PINK YELLOW BATCH NO. (2)

FRONT

The card displays a grid of fabric swatches. A diagonal line is drawn across the grid. Handwritten notes include circled numbers 1, 2, and 1P, and the text '=> REV' and '=> CREAM SOLID'. A signature is visible at the bottom right.

=> REV

=> CREAM SOLID

KTF-482

Figure 3.15: Continuity Card

3.6 Quality

The main purpose of quality checker is to check the quality of the process of cutting being. There are various standards of quality on which quality inspector mention that this

particular part is quality passed. Those standards is simply the CM: Contract for made ups. Quality standard also includes that size, colour and dimensions are upto the mark. Quality is also set according to the type of cloth that either it is upto the mark that the contrast, embroidery is done upto the mark or not and also the quality of cut is checked visually by quality checkers.

3.7 Layering process

Layering of the fabric is done both manually and automatically.

Layering of the fabric is done for the several reasons:

1. Efficient Cutting
2. Reduce Waste
3. Accuracy
4. Improved Productivity
5. Cost Effective

For different fabric, the thickness of layer must be in consideration so that the layers can be cut properly. So those thickness are different for different fabrics:

1. Sateen (3-4 inch)
2. Percale (5-6 inch)
3. Flannel (7-8 inch)

3.8 Defects of Cutting

1. Wrong Direction Cutting
2. Uneven Cutting (Rap)
3. Size Short
4. Cloth Shrinkage

3.9 Resources of Cutting Department

There are different resources of cutting department. The main function of cutting department is to cut the fabric according to the design. Which includes:

1. Machine
2. Labor
3. Cutting Plan
4. Table
5. Article
6. Customer
7. Product

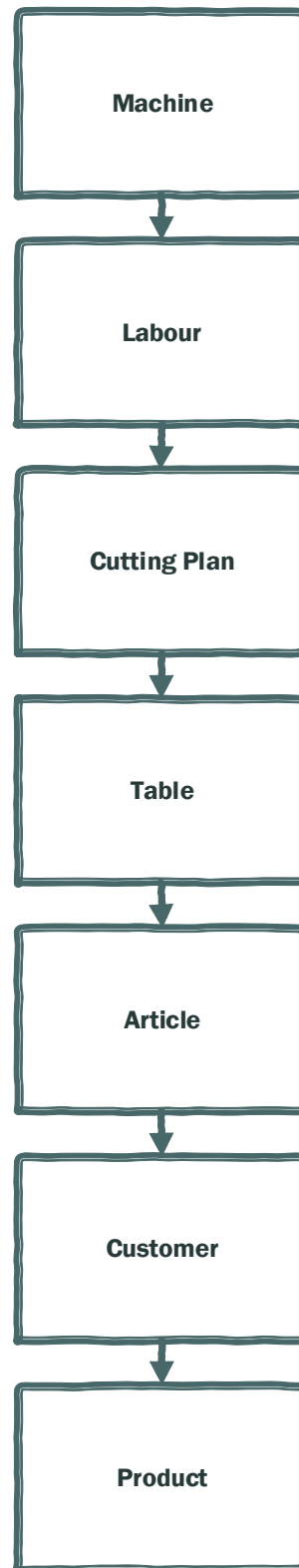


Figure 3.16: Resources of Cutting Department

Chapter 4: Stitching

4.1. Definition

The stitching department work is to stitch the upcoming cloth from cutting and forward that to packing area. In stitching department there are various stitching machines that stitch the cloth and checker checks the production of the parts of stitching cloth and on the checking table the checker check the shortcomings if occurred in the stitching.

4.2. Staff Hierarchy

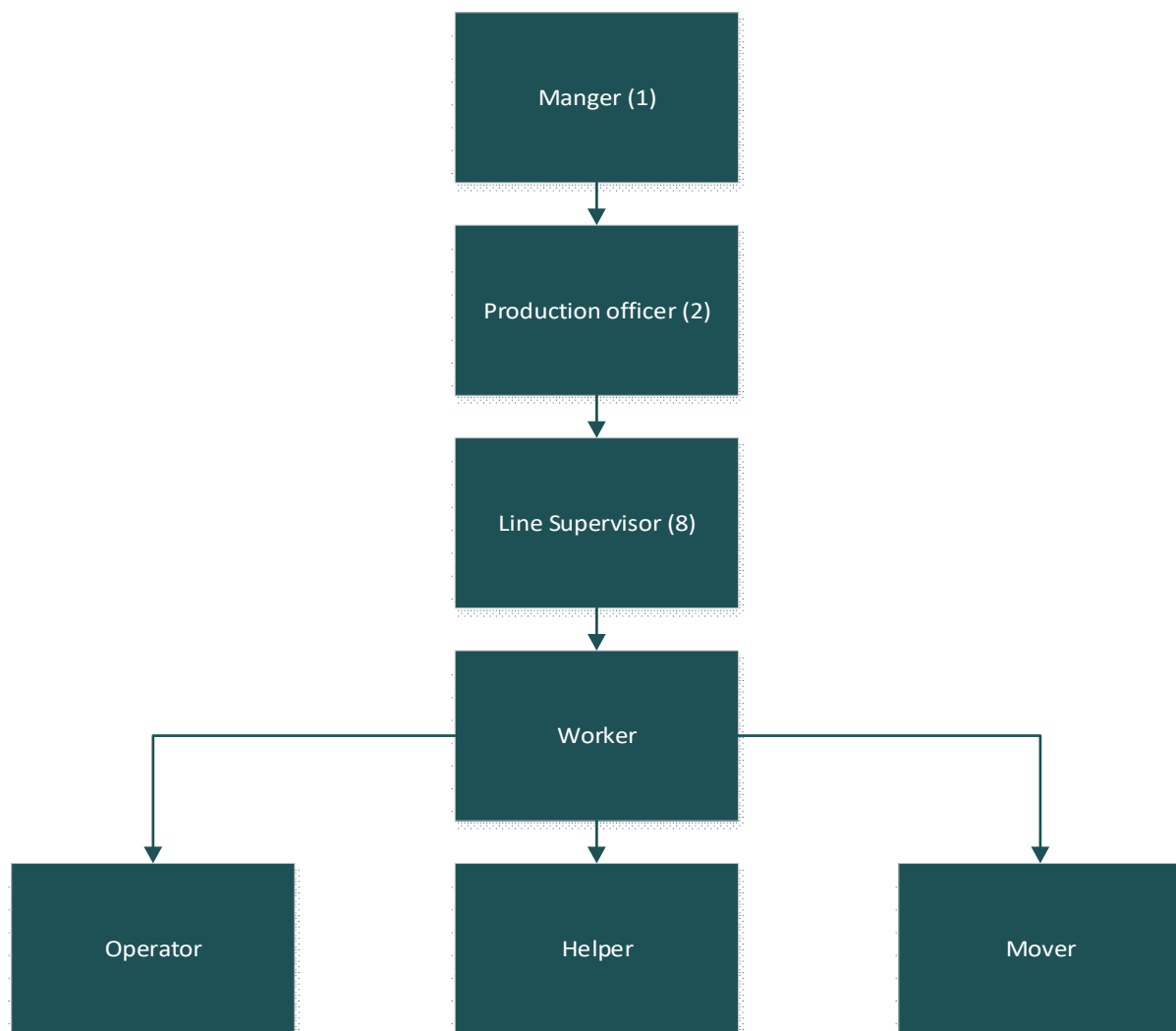


Figure 4.1: Staff Hierarchy

Staff Hierarchy is related to the staff working under the hierarchy from top to bottom which means from Manger to Workers of stitching staff.

Table 2: Staff Quantity

Staff	Quantity
Manager	1
Production Officer	2
Line Supervisor	8
Workers	(70+30)

4.3 Stitching Section Hierarchy

Layout of the stitching unit is based up on the units and lines under which the machines and workers are working for the production of the stitching articles:

Stitching has **four units**. Every unit is producing different articles based on the CM number.

But their layout is almost same that contains:

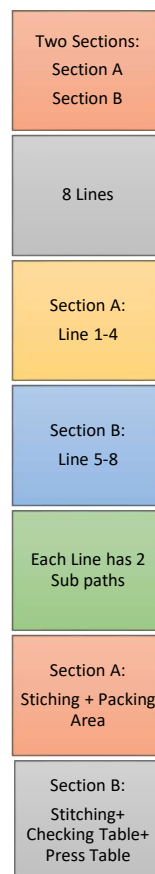


Figure 4.2: Stitching Section Hierarchy

4.4. Process Flow

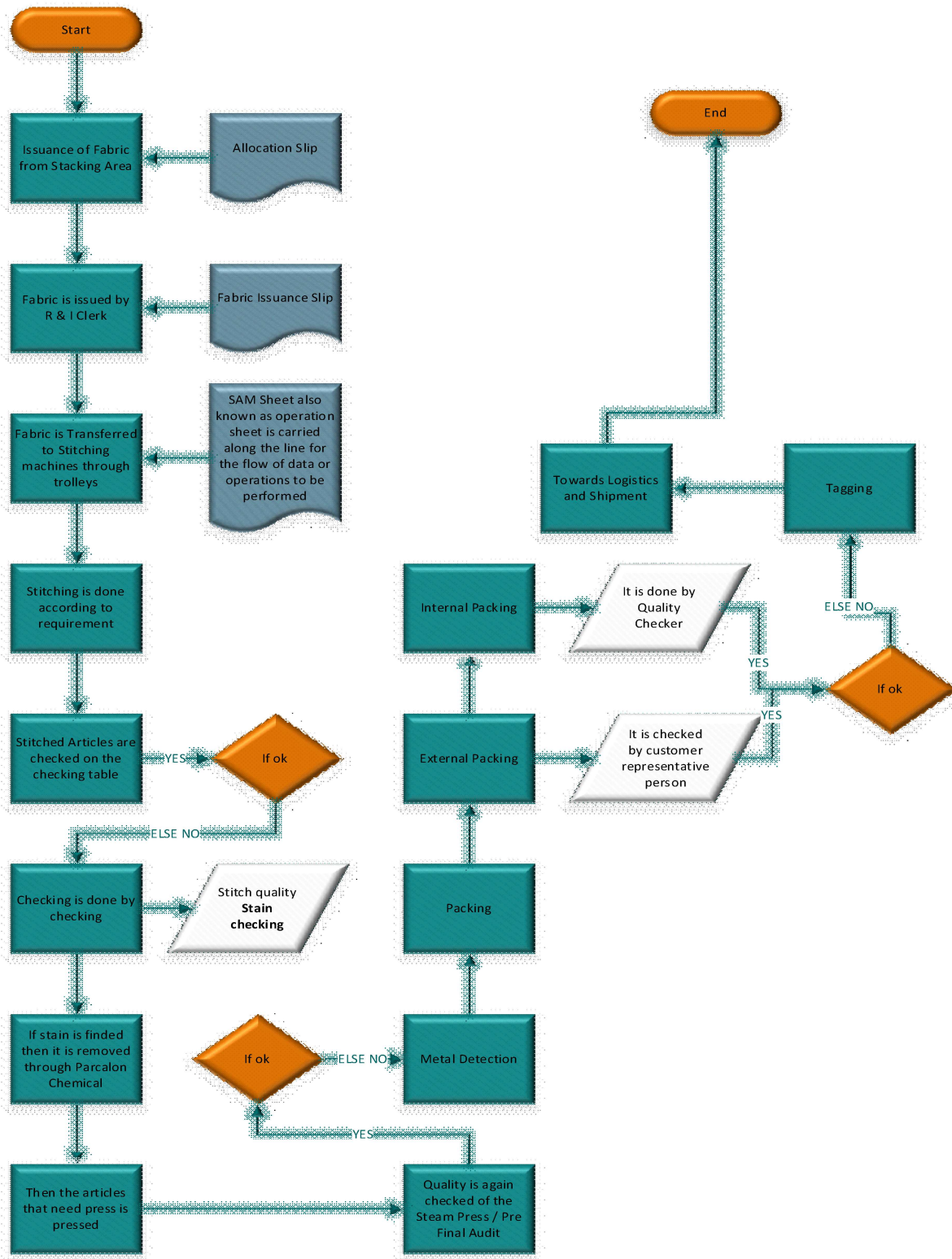


Figure 4.3: Stitching Process

Our process flow goes in a step by step manner that firstly our fabric is issued from the stacking area through the allocation slip and R & I clerk bring that fabric.

Then the fabric is transferred to stitching machine area, which are mainly divided into 8 lines through trolleys. Large set of articles like bedsheet, fitted and quilt cover are hanged up on the chains while small articles like pillows are carried through tables. The stitching is completed their and then stitched articles are transferred to checking area.

In checking different things are checked which mainly includes stain, stitched quality.

After checking articles are send to the washing area if founded with the stain and the stains are removed through the chemicals which has the composition of water and chemical parcalon.

Then the articles are again checked that the stains are either removed properly or not with the chemicals.

Articles that need the press are pressed accordingly and their press quality is checked.

Then the articles are detected with metal through metal detecting machine and are finally packed with the company logo or tag.

4.5 Stitching Machines

There are various types of stitching machines in the Stitching units that are used for the various types of the stitches based on the type of the requirement an article demands.

So those stitching machines are:

4.5.1 Singer

Singer is a single needle cutting machine that is used for the articles that requires the single needle stitch for the stitching process. The singer uses 2 threads, 1 bobbin and 1 thread for the stitching process. The lock stitch is formed when the needle thread and bobbin thread lock each other.



Figure 4.4: Singer

4.5.2 Double Needle Stitching Machine

A double needle stitching machine is a sewing machine that uses two needles for stitching the fabrics. It is normally used to stitch the hem on to the fabrics or for designing purpose. In this the

stitching is done in a way that it sew two parallel lines on the top of the fabric while the bobbin thread zigzags between the stitches on the underside of the fabric.

The sewing technique of this fabric allows customer to get high quality of stich on the fabric that is reliable and strong.

It uses one bobbin thread and two needles that are connected by a shank.



Figure 4.5: Double Needle Stitching Machin

4.5.3 Overlock Stitching Machine

An overlock-stitching machine is a type of stitching machine that is used to give high strong professional stitch to the fabric edges. It uses two or four threads depending upon the type of stitch used. An overlock machine uses two needles that stitch the parallel lines and two or four looper threads that interlace at the edge of the fabric making a tear resistive stitch with a strong bond of stitch.

Table 3: Over Lock Types

Over Lock Types	Looper	Needles
2 Thread	1	1
3 Thread	1	2
4 Thread	2	2
5 Thread	2	3



Figure 4.6: Overlock Stitching Machine

4.5.4 Baratta Stitching Machine

It is a type of sewing machine that uses a combination of two needles and two looper thread to create a strong and stretchy seam. It is used to create a professional finish.

Actually in this the two needle sew the parallel lines while the two looper thread interlace at the edges of the fabric creating a durable seam.



Figure 4.7: Baratta Stitching Machin

4.5.5 Bar Take Stitching Machine

A Bar Take sewing machine is a type of sewing machine that is used to create more stitches repeatedly at on place to create the strength of stitch at that area so that the fabric near that area can bear the load. This sewing machine is only made to stitch at small area to create the strength at that area.

It has many of the purposes like:

1. Attaching belt loops
2. Increasing strength in the corner of a pocket
3. Closing the two corner of the button hole
4. At the end of the zip
5. At any place where more strength is needed to bear extra load

The Bar Take machine uses 1 needle thread and 1 bobbin thread.



Figure 4.8: Bar Take Stitching Machine

4.6 Pressing Table

The articles that need press are pressed accordingly in the press area on the pressing table.

The press is done using the steam-ironing machine that is so satisfying because the quality of press through this machine is phenomenal.

A steam iron machine use the phenomenon of steam to remove wrinkles and creases from fabric.

There are 3 Press Table and 9 ironing machines.



Figure 4.9: Steam Iron Stitching Machine

4.7 Checking Table

On checking table, different things checked regarding the stitch quality, hem size, stains etc.

If the fabric founded with the stain then its send to the washing table where the help of the chemical named **parcalon** removes the stains from the fabric with the mixture of water.

Checking table is helpful in checking the flaws and by revising the issues in the flaws the flaws are overcome.

4.8 Packing Area

The articles after having stitching and other operations are send to the packing area where metal free fabric is packed in a professional way with premium packing quality and is packed in air tight boxes and are marked with tags on which all the details regarding brand, area, quantity in box is mentioned. So the packing is final stage of the process so its quality is tried to be set up to the mark.

The article set for packing checked accordingly as:

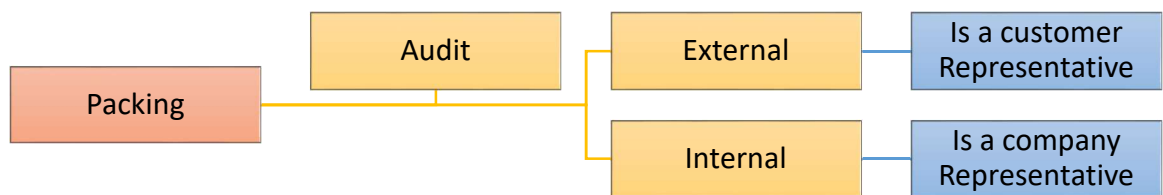


Figure 4.10: Packing Audi



Figure 4.11: Packing Area

4.9 Types of Stitches

1. Straight Stitch
2. Zig zag Stitch
3. Overlock Stitch
4. Blind Hem Stitch

4.10 Stitching Defects

1. Needle Hole
2. Stain
3. Wrong Label
4. Grainage
5. Thread Alter

4.10.1 Needle Hole:

A needle hole is a defect in stitching refers to an unwanted hole in the fabric, caused by the needle in the sewing process. This can occur due to various reasons such as incorrect needle size, improper stitching technique, poor fabric, needle damage, sewing machine issue etc

4.10.2 Stain

It is a common issue in the stitching process, be either caused by the machine oil, or due to any other issue during handling and this stain can be easily removed by the chemical parcalon.

4.10.3 Wrong Label

It refers to the mistake in the labelling such as incorrect text, wrong logo, misplaced labels.

The defect can mainly occur due to different reasons such as Human error, design issues, equipment problems, material issues.

Wrong label can lead to production rejection, delay in production, loss of customers trust.

To minimize this issue double check design, use high quality material and equipment, train operator and designer, implement quality control check.

4.10.4 Grainage

Grainage refers to the direction or alignment of the fiber. It can effect fabric stability, stitch quality, fabric durability, print or dye uniformity.

Grainage is classifies into three types warp grainage, weft grainage, bias grainage.

4.10.5 Thread Alter

It is a defect in stitching in which thread deviates from its intended path causing the alteration in the stitch pattern. This can result in an uneven stitch, distorted fabric, visible thread loop, inconsistent tension.

4.11 NCP

It refers to products i.e. goods and services that fail to meet specified requirements, standard or customer expectations.

1. NCP (Non-Conforming Products) can be either:
2. Faulty Products
3. Incorrect Labelling
4. Fails to meet safety Standards
5. Poor Quality or Performance of Products

Chapter 5: Embroidery

5.1 Definition

Embroidery is the art and technique of decorating the fabric with different designs. Those designs either can be in small design patches or in the form of long detailed designed.

These designs are firstly designed on the software according to the requirement of the customer and is then transferred to the Embroidery software system via USB for transmitting that design on to the fabric.

With that design all the details regarding the size, colors of the embroidery and other minor details are also transferred with that design.

5.2 Design Process Flow

There is a complete hierarchy of the flow of the design that can be understandable by the flow diagram given below:

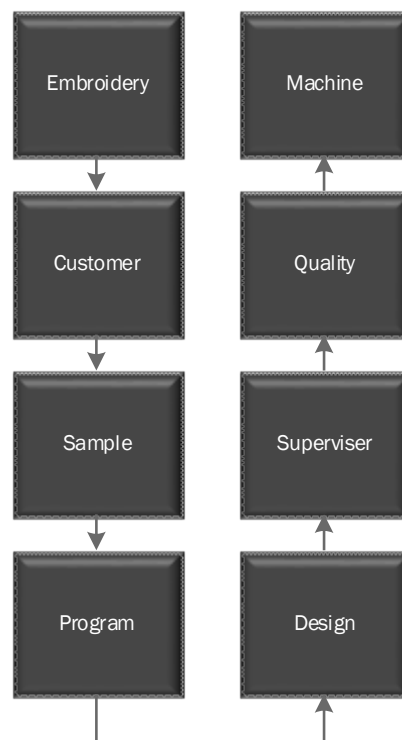


Figure 5.1 Design Flow

5.3 Staff Hierarchy

The staff works in the embroidery section that can be understandable by the flow diagram in a structure:

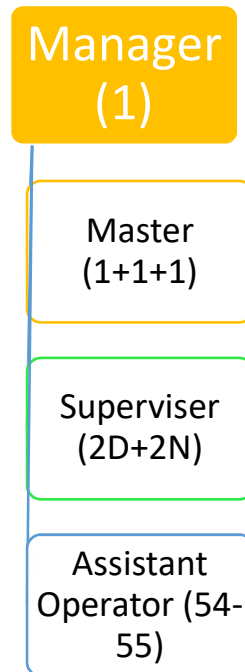


Figure 5.2 Staff Hierarchy

5.4 Machines

5.4.1 Number of Halls

In total there are 3 Halls of Embroidery which are mainly used for different kind of embroideries having a little different specifications from each other.

5.4.2 Specs of Embroidery Machines

Table 4: Specs of Embroidery Machines

	H to H Distance	Needles	Bobbins	# of Heads	Pressure Boot	RPM	Frame Length	Frame Width
ZSM	16 in	9	1	28	9	1000	11.3 m	1800 mm
Chenille	18.5 in	9	Has Thread Roll instead of Bobbin	24	9	1000	11.3 m	1800 mm
SWM	13 in	9	1	24	9	900	5 m	1800 mm

Sample Machine:

Head to Head Distance: 39 in

Frame Length: 8 m

RPM: 1000 RPM

Frame Width: 1800 mm

5.4.3 Details of Machine

5.4.3.1 SWM

An Embroidery SWM is a computerized Embroidery Machine used for embroidery design on fabric.

It has high precision and Accuracy.

It has Adjustable stitch length and width.

It is suitable for small to medium size designs.



Figure 5.3: SWM Machine

5.4.3.2 ZSM

ZSM Embroidery machine is used to perform high speed computerized embroidery on fabrics.

It is used to create designs and Patterns with precision and accuracy.

Increase efficiency and productivity in embroidery operations.



Figure 5.4: ZSM Machine

5.4.3.3 Chenille Embroidery Machine

Chenille embroidery machines are designed to perform chenille embroidery using loop and chain stitches offering various features and benefits:

1. Unique Texture
2. Optional Features
3. High Speed Operation
4. Automatic Design
5. Compact Design



Figure 5.5: Chenille Embroidery Machine

5.4.3.4 Winder Machine

It is an automated machine used for filling the bobbin with the thread. It only needs the thread attached to it and a bobbin that is placed on the bobbin area and then in a few seconds it fill the bobbin with the thread.



Figure 5.6: Winder Machine

5.5 Types of Stitches

Following are some common types of stitches that are normally used in the embroidery:

1. Sartan
2. Tatami
3. Zigzag
4. E-stitch
5. Motif
6. Contour
7. Program Split

5.6 Common Embroidery Defects:

Following are some common embroidery defects:

1. Bite Variation
2. Applique Out
3. Skip Stitch
4. Design Out
5. Missing
6. Bobbin Thread Shown
7. Shade Variation
8. Black Stain
9. Loose Stitch
10. Embroidery Damage

5.6.1 Bite Variation

In embroidery bite variation is occurs when the distance from one stitch to another stitch changes and does not remains the same throughout the stitch. It leads to the customer rejection and improper design.

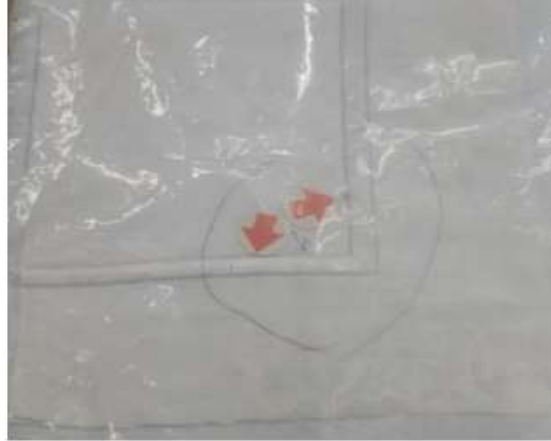


Figure 5.7: Bite Variation

5.6.2 Loose Stitch

It is the defect in which stitch size become greater than the normal stitch. This issue causes the defect in the design and ultimately leads to customer loss and order rejection. This defect can be eliminated by using the proper stitch size according to the requirement.



Figure 5.8: Loose Stitch

5.6.3 Black Stain

It is mainly caused due to the stain caused during dye or due to the machine oil. This issue can be resolve by washing but increase the production time.



Figure 5.9: Black Stain

5.6.4 Bobbin Thread Shown

This issue is caused when the bobbin thread comes on with upper embroidery, which causes the color difference in the embroidery design. This is a big issue as the thread of bobbin is of different colour sometimes because the fabric is two toned sometimes as its front is of some colour and back is of some so if the back color bobbin thread comes on the front embroidery it can causes to a big problem.



Figure 5.10: Bobbin Thread Shown

5.6.5 Shade Variation

This issue mainly arises if the thread recommended by the customer for the particular colored fabric differs or changed due to mistake can cause shade difference between embroidery and fabric color.



Figure 5.11: Shade Variation

5.6.6 Applique Out

It is caused due to the addition of the extra fabric onto the main fabric and this issue mainly arises if the framing of the fabric is not done properly onto the frame. So this issue can be eliminated if the fabric is attached properly onto the frame.



Figure 5.12: Applique Out

Chapter 6: Quilting

6.1 Definition

Quilting is the process in which a quilt is made by warding and through stitch process in which polyester or cotton is added in the covering to make quilt.

Quilt is made through either quilt process or from Delta machine in which the quilt is made through stitch to fix the cotton or polyester in the quilt cover.

6.2 Types of machine in quilting Section

In total, there are three types of machines in the quilting section, which have its following quantity:

Table 5: Types of Machine In Quilting

Machines	Quantity
Delta Machine	13
Wadding Machine	1
Stitching Machine	4

6.3 Delta Machine Works

The work of Delta Machine is to stitch the cotton or polyester warded quilt to make it fix in quilt cover to make quilt cover according to the computerized input.

It works on the SMED process to make the work in progress (WIP) fast and in a flow.

So it has one head that moves in one direction to make a stitch and a moving table that moves in alternative direction to make a stitch.

As on quilt is stitched then immediately the worker adds other frame of quilt so next quilt can be stitched to make the work in progress fast and flow.

It has one Bobbin and one needle.



Figure 6.1 Delta Quilting Machine

6.4 Wadding Filling Machine

The cotton or polyester is added in the quilt cover to make the quilt in the process.

It is the semi-automated process in which a worker puts the quilt cover on two rods and the machine puts the cotton or polyester in the quilt cover.

Two rods help to create the space for the cotton to come in the quilt cover.



Figure 6.2: Wadding Filling Machine

6.5 Stitching Machine

The one open side after wadding of the quilt is stitched which is done by the help of stitching machine.

6.6 Defects in Quilting

Following are some of the defects of Quilting:

1. Skipped Stitches
2. Inconsistent Stitches
3. Insufficient filling of wadding
4. Loose Stitch
5. Stain

Chapter 7: Multi Needle Embroidery

7.1 Definition

In the process of Embroidery, Embroidery is done in the straight line by moving or passing the fabric roll through roller from the needles to do embroidery on to the quilts.

7.2 Staff Hierarchy

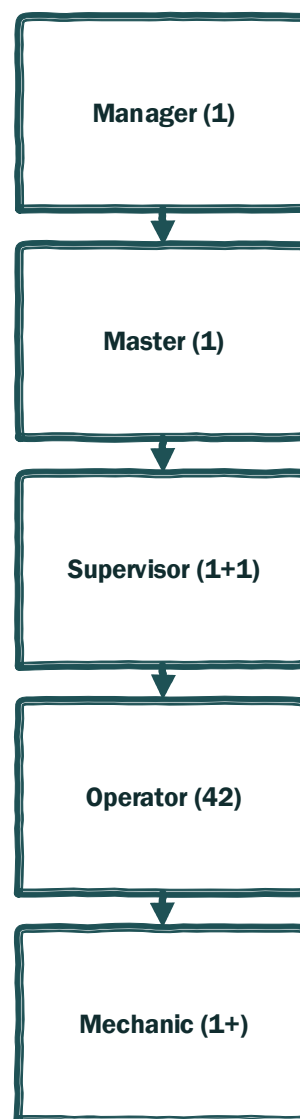


Figure 7.1: Staff Hierarchy

7.3 Machines

There are 22 Machines:

1. (19) Multi needle
2. (1) ABM Alteration
3. (2) Heing-Ye

7.3.1 Multi Needle Embroidery

The machine is used to stitch two fabrics having wadding in it. It has roller which rolls the fabrics and allow it to pass through the needles which are helpful in doing embroidery on it. It has 32 heads containing 64 needles and 64 bobbins. It has the capability to stitch the fabric filled up to 250 GSM of Polyester and 220 GSM of cotton. In this the needle to needle distance 2 inch and width of the machine is 180 mm.



Figure 7.2: Multi needle Embroidery

7.3.2 ABM Alteration Machine

It is a machine that is used to fill the missed gaps of the other machines, it is helpful in making the stitch of Embroidery completely which are missed during the embroidery process. It is helpful in bringing exactly the same stitch that are in the flow previously. So from where the stitch is skipped by the machine, this machine brings back that missed stitch.

7.3.3 Multi-Needle Heing Ye

It is a type of machine that is used to create the stitch in a straight-line embroidery. It has rollers on which that fabric is loaded and then is passed through the 300 needles on equal interval to create the embroidery. This machine has 5 rollers and instead of bobbin it has ship on which thread is loaded. Ambry is used on the roller for the grip of the fabric.



Figure 7.3: Multi-Needle Heing-Yeh

7.4 Defects

1. Needle Break
2. Grain age
3. Oil Stain
4. Shuttle Out

7.5 Design

1. Channel Quilt
2. Verma Chilli
3. Diamond
4. Cable Quilting
5. Cloud
6. Goggi

7.6 Difference between Embroidery and Multi-Needling

The vital difference between Embroidery and Multi –Needling is the difference of mechanism that Embroidery uses the frames while Multi-Needling uses the rollers. So in Embroidery the fabric is loaded on to the frame and frame moves in x and y directions to do embroidery on the fabric. While in Multi-Needling fabric is loaded on the rollers and rollers help the fabric to move from the needles from where the embroidery takes place.

Chapter 8: Supply Chain

8.1 Definition

A supply chain is a flow of Material from procurement to client, which involves different processes through which material flow in a hierarchy. The main process take place in any industry for the flow of goods and services. It is the complete process of flow in which documentation is kept for each process for keeping the record of every process.

8.2 Process Flow



Figure 8.1: Process Flow

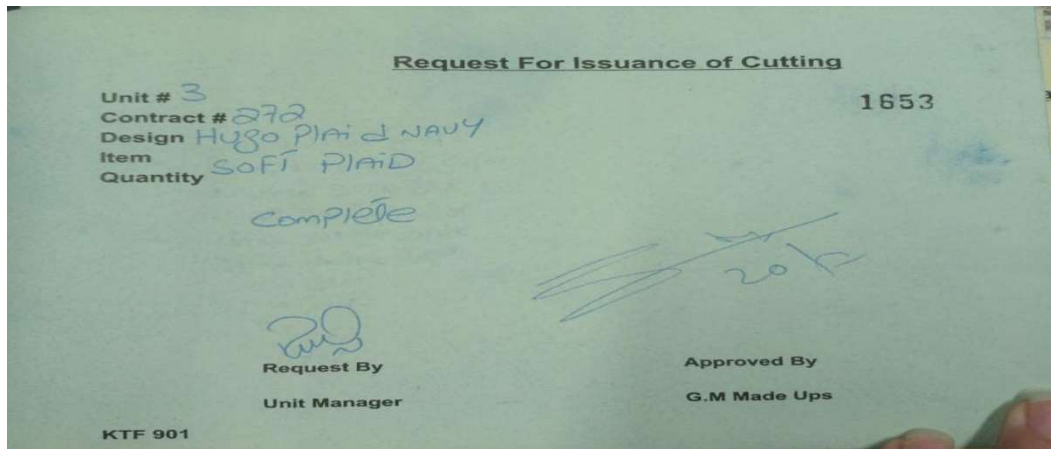
A supply chain is a network of organization, people, activities, and resources involved in the production and delivery of a product or service from a raw material stage to the end customer.

Chapter 9: Documentation Flow of Stitching

We will discuss the documentation flow of stitching department from issuance of cutting from cutting go down to warehouse.

9.1. Allocation Slip

Allocation slip is also called the request for issuance of cutting slip. Line supervisor make the allocation slip and then R and I clerk took it to the cutting go down. Allocation slip is approved by the clerk, manager and the DGM of made-ups. Original allocation slip is taken by the stitching department after the fabric issuance and its copy is taken by the cutting go down. In this report, unit # , contract #, design, item and quantity is mentioned.



Request For Issuance of Cutting

Unit # 3
 Contract # 272
 Design HUGO PLAID NAVY
 Item SOFT PLAID
 Quantity COMPLETE

1653

Request By [Signature]
 Unit Manager

Approved By [Signature]
 G.M Made Ups

KTF 901

Figure 9.1: Allocation Slip

9.2. Cutting Issue Book

When the R & I clerk reach the cutting go down and the counting and bundling officers enters all the details from allocation slip to cutting issue book. In cutting issue book date and shift is mentioned on the top. Then the CM # , design , version , article , size , no of packing , pieces, cut size is mentioned vertically. While the signatures of prepared by, checked by and received by is mentioned on the bottom. It has 2 copies one is taken by the R & I Clerk and one is kept by the cutting go down.

(Made Ups)

Page No. 11

Job No. _____

CM No. 183

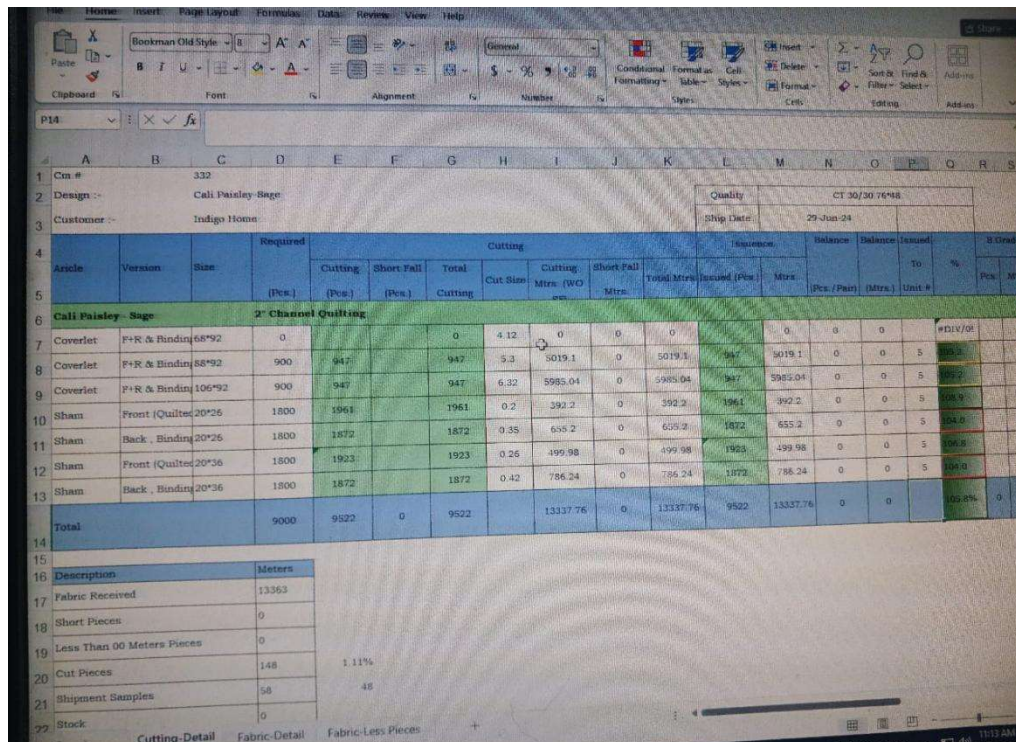
Construction / Quality: _____

Cutting Pieces	Issued Unit #	Issued Pieces
Layer Wise Cutting Pcs $\frac{18}{10} \frac{15}{10} \frac{10}{10}$ $57 + 12 + 14$ $\frac{24}{24} \frac{28}{28} \frac{47}{47}$	2 A	2201 2215 $57+12+14$ $5/2$ 6/2
$\frac{18}{10} \frac{15}{10} \frac{10}{10}$ $57 + 12 + 14$ $\frac{24}{24} \frac{28}{28} \frac{47}{47}$	2 A	2201 2215 $57+12+14$ $5/2$ 6/2
$\frac{18}{10} \frac{15}{10} \frac{10}{10}$ 130 $\frac{24}{24}$	2 A	2201 130 $5/2$
$\frac{18}{10} \frac{15}{10} \frac{10}{10}$ 130 $\frac{24}{24}$	2 A	2201 130 $5/2$
$\frac{9}{10} \frac{38}{10} \frac{76}{10} \frac{4}{10}$ $30+164+10+6+6$ $\frac{24}{24} \frac{28}{28} \frac{117}{117} \frac{210}{210} \frac{47}{47}$	2 A	2201 2215 $30+164+10+6+6$ $5/2$ 6/2
$\frac{9}{10} \frac{38}{10} \frac{76}{10} \frac{4}{10}$ $30+164+10+6+6$ $\frac{24}{24} \frac{28}{28} \frac{117}{117} \frac{210}{210} \frac{47}{47}$	2 A	2201 2215 $30+164+10+6+6$ $5/2$ 6/2
$\frac{76}{10} \frac{76}{10} \frac{4}{10}$ $92 + 5 + 3$ $\frac{24}{24} \frac{28}{28} \frac{117}{117} \frac{100}{100}$	2 A	2201 $92+5+3$ $5/2$
$\frac{76}{10} \frac{76}{10} \frac{4}{10}$ $92 + 5 + 3$ $\frac{24}{24} \frac{28}{28} \frac{117}{117} \frac{100}{100}$	2 A	2201 $92+5+3$ $5/2$

Cutting Delivery Incharge

Figure 9.2: Daily Cutting Ledger-2

After entering the data on daily cutting ledger, computer operator enter all the data from daily cutting ledger on the computer on excel sheet.



Article	Version	Size	Required	Cutting	Short Fall	Total	Cut Size	Cutting Mtrs (WO)	Short Fall Mtrs	Total Mtrs	Balance	Balance Issued	Balance To	%	Pcs	Mtrs
Coverlet	F+R & Binding	68*92	0	0	0	4.12	0	0	0	0	0	0	0	0	0	0
Coverlet	F+R & Binding	68*92	900	947	0	947	6.32	5985.04	0	5985.04	947	5985.04	0	0	0	0
Coverlet	F+R & Binding	106*92	900	947	0	947	6.32	5985.04	0	5985.04	947	5985.04	0	0	0	0
Sham	Front (Quiltes)	20*26	1800	1961	0	1961	0.2	392.2	0	392.2	1961	392.2	0	0	0	0
Sham	Back, Binding	20*26	1800	1872	0	1872	0.35	655.2	0	655.2	1872	655.2	0	0	0	0
Sham	Front (Quiltes)	20*36	1800	1923	0	1923	0.26	499.98	0	499.98	1923	499.98	0	0	0	0
Sham	Back, Binding	20*36	1800	1872	0	1872	0.42	786.24	0	786.24	1872	786.24	0	0	0	0
Total			9000	9822	0	9822		13337.76	0	13337.76	9822	13337.76	0	0	0	0

Figure 9.3: Data on Excel Sheet

9.4. Quality in Line Inspection Report

During stitching process in line inspection is done by the quality inspector in 4 rounds. In these 3 rounds 2 is done before break and 2 is done after the break. Quality inspector check the pieces of every operator and then write the defects on the report. In line inspection report , CM #, customer name, QC inspector, production supervisor, line 3 and floor is mentioned on the top. And machine no operation , design and color defects details and operation specs is mentioned vertically. Quality in line inspection report is approved by the quality inspector, production supervisor, manage stitching and Manager QC.

9.7. Pre Final Inspection Report

It is a computer generated report.

1202

KOTML
 Date: 31/12/24
 Cm: 005-23
 Customer: Lunatic
 Design: Future
 Color: Multi
 Item: Duvet
 Size: Mix
 Req.Q.

AQI 1.5/2.5/10/11

Made up Quality
 Pre-Final Inspection Report
 Unit: 02A
 Auditor: N. N. N.

Time	F/S Lot	Status	Pass Quantity	Pack Status	Fault Detail	Minimum Size	Maximum Size	Checker Name	Supervisor Name
					(270x240)				
13	62	Rej			slant label and label whitening show	267x237	270x241		
					Border Variation	267x237	270x241		
					Unseen Mark	267x237	270x241		
					Mark Variation	267x237	270x241		
13	60	Pass	60		Unseen Mark	267x237	270x241		
					(245x210)				
20	100	Pass	100		Border Variation	244x209	245x211		
					slant label	244x209	246x212		
					(270x240)				
13	63	Pass	63		Unseen Mark	267x237	270x241		
					Border Variation	267x237	270x241		
					label whitening show	267x237	270x241		
09	59								
285	283								
Total									

Production Manager / Officer
 KTF-877

Quality Officer

Assistant / Manager Quality

Figure 9.9: Pre Final Inspection Report

9.8. Daily Quality Performance Report

It is a computerized report of all the units of the stitching. In this report, there are two sections, one is in line inspection summary and second is pre final inspection summary. In in line inspection summary, daily in line inspection percentage, current week in line defects %, monthly line wise in line defects %, last 6 months defects %, last 6 months top 5 defects is mentioned. And in pre final inspection summary, daily pre final inspection, current week pre final defects %, monthly pre final defects %, monthly failure % and 6 months top 5 defects is mentioned.

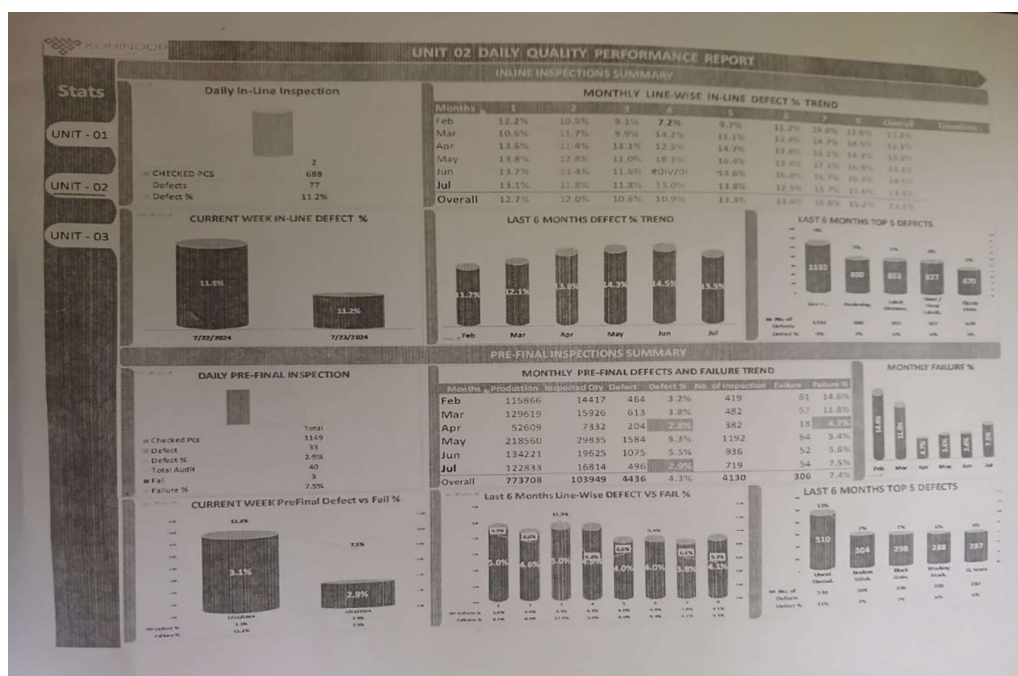


Figure 9.7: Daily Quality Performance Report

9.9. B-Grade Fabric Report

B grade fabric report is prepared at the end of the day. In this report CM#, design, color, version, party name, item, size, order, cutting, packing, total damage, and reason of damage in folding and made ups is mentioned. B grade fabric report has 4 copies, 1 is send to the quality, 1 send to the order closing area, 1 is kept by the unit, and 1 is send to the local stitching. In the end, it is approved by the marketing manager, GM made ups, unit manager, unit incharge, manager PPC, and quality officer. After the approval, stitch fabric with defects is send to the local stitching and the un stitched fabric is send back to the folding.

9.11. Final Inspection Report

Final inspection report is prepared after the final audit. It is a computerized report. In this report CM#, customer, unit, inspection date is mentioned at the top. And PO#, item, size, color/design, required quantity, prepared quantity, packing sample and full quantity is mentioned vertically.

FINAL INSPECTION REPORT											
CM #	24-180	UNIT #	3	SECTION	518	DEPT. OFFICER	107				
CUSTOMER	Pranveer	REQ. QTY.	4147	REAL QTY.	4147	INSPECTOR	Pranveer				
INSPECTION DATE	27-Jun-24	INSPECT QTY.	4090	TOTAL QTY.	4147	INSPECTOR NAME	Pranveer				
		MAJOR QTY. %	98.4%			PREPARED BY	Pranveer				
SAMPLING PLAN		ACCEPTANCE / REJECTION CRITERIA		SAMPLE SIZE		MODELS					
Inspection Type	Final	Acceptance	98.4%	Rejection	1.6%	Sample Size	100				
Inspection Level	Final	Acceptance	98.4%	Rejection	1.6%	Sample Size	100				
Inspection Method	Visual	Acceptance	98.4%	Rejection	1.6%	Sample Size	100				
PACKAGING INSPECTION		WORKMANSHIP INSPECTION		OVERALL RESULT							
TOTAL INSPECTED QTY.	4090	TOTAL DEFECTIVE QTY.	65	PASS							
MAJOR DEFECT %	1.6%	MAJOR DEFECT %	1.6%								
MAJOR DEFECT %	1.6%	MAJOR DEFECT %	1.6%								
ITEMS DESCRIPTION											
SL. NO.	PO #	ITEM	SIZE	COLOR / DESIGN	REQ. QTY.	PREPARED QTY.	PACKING SAMPLE	QTY PUT UP	NO. OF QTY.	REAL QTY.	WORKMANSHIP
1	007-004	Shower	45x70	Antique Teak	120	120	8	34	1	1	8
2	007-004	Shower Set	130x200x100mm	Antique Teak	30	30	2	10	1	1	8
3	007-004	Shower Set	130x200x100mm	Antique Teak	30	30	2	10	1	1	8
4	007-004	Shower Set	130x200x100mm	Antique Teak	30	30	2	10	1	1	8
5	007-004	Shower Set	130x200x100mm	Antique Teak	30	30	2	10	1	1	8
6	007-004	Shower Set	130x200x100mm	Antique Teak	30	30	2	10	1	1	8
7	007-004	Shower Set	130x200x100mm	Antique Teak	30	30	2	10	1	1	8
8	007-004	Shower Set	130x200x100mm	Antique Teak	30	30	2	10	1	1	8
9	007-004	Shower Set	130x200x100mm	Antique Teak	30	30	2	10	1	1	8
10	007-004	Shower Set	130x200x100mm	Antique Teak	30	30	2	10	1	1	8
11	007-004	Shower Set	130x200x100mm	Antique Teak	30	30	2	10	1	1	8
12	007-004	Shower Set	130x200x100mm	Antique Teak	30	30	2	10	1	1	8
13	007-004	Shower Set	130x200x100mm	Antique Teak	30	30	2	10	1	1	8
14	007-004	Shower Set	130x200x100mm	Antique Teak	30	30	2	10	1	1	8
15	007-004	Shower Set	130x200x100mm	Antique Teak	30	30	2	10	1	1	8
16	007-004	Shower Set	130x200x100mm	Antique Teak	30	30	2	10	1	1	8
17	007-004	Shower Set	130x200x100mm	Antique Teak	30	30	2	10	1	1	8
18	007-004	Shower Set	130x200x100mm	Antique Teak	30	30	2	10	1	1	8
19	007-004	Shower Set	130x200x100mm	Antique Teak	30	30	2	10	1	1	8
20	007-004	Shower Set	130x200x100mm	Antique Teak	30	30	2	10	1	1	8
21	007-004	Shower Set	130x200x100mm	Antique Teak	30	30	2	10	1	1	8
22	007-004	Shower Set	130x200x100mm	Antique Teak	30	30	2	10	1	1	8
23	007-004	Shower Set	130x200x100mm	Antique Teak	30	30	2	10	1	1	8
24	007-004	Shower Set	130x200x100mm	Antique Teak	30	30	2	10	1	1	8
25	007-004	Shower Set	130x200x100mm	Antique Teak	30	30	2	10	1	1	8
26	007-004	Shower Set	130x200x100mm	Antique Teak	30	30	2	10	1	1	8
27	007-004	Shower Set	130x200x100mm	Antique Teak	30	30	2	10	1	1	8
28	007-004	Shower Set	130x200x100mm	Antique Teak	30	30	2	10	1	1	8
29	007-004	Shower Set	130x200x100mm	Antique Teak	30	30	2	10	1	1	8
30	007-004	Shower Set	130x200x100mm	Antique Teak	30	30	2	10	1	1	8
31	007-004	Shower Set	130x200x100mm	Antique Teak	30	30	2	10	1	1	8
32	007-004	Shower Set	130x200x100mm	Antique Teak	30	30	2	10	1	1	8
33	007-004	Shower Set	130x200x100mm	Antique Teak	30	30	2	10	1	1	8
34	007-004	Shower Set	130x200x100mm	Antique Teak	30	30	2	10	1	1	8
35	007-004	Shower Set	130x200x100mm	Antique Teak	30	30	2	10	1	1	8
36	007-004	Shower Set	130x200x100mm	Antique Teak	30	30	2	10	1	1	8
37	007-004	Shower Set	130x200x100mm	Antique Teak	30	30	2	10	1	1	8
38	007-004	Shower Set	130x200x100mm	Antique Teak	30	30	2	10	1	1	8
39	007-004	Shower Set	130x200x100mm	Antique Teak	30	30	2	10	1	1	8
40	007-004	Shower Set	130x200x100mm	Antique Teak	30	30	2	10	1	1	8
41	007-004	Shower Set	130x200x100mm	Antique Teak	30	30	2	10	1	1	8
42	007-004	Shower Set	130x200x100mm	Antique Teak	30	30	2	10	1	1	8
43	007-004	Shower Set	130x200x100mm	Antique Teak	30	30	2	10	1	1	8
TOTAL					4147	4090	200	400	100	200	
ITEM WITH MAJOR DEFECTS											
1	007-004	Shower	45x70	Antique Teak	120	120	8	34	1	1	8
2	007-004	Shower Set	130x200x100mm	Antique Teak	30	30	2	10	1	1	8
3	007-004	Shower Set	130x200x100mm	Antique Teak	30	30	2	10	1	1	8
4	007-004	Shower Set	130x200x100mm	Antique Teak	30	30	2	10	1	1	8
5	007-004	Shower Set	130x200x100mm	Antique Teak	30	30	2	10	1	1	8
6	007-004	Shower Set	130x200x100mm	Antique Teak	30	30	2	10	1	1	8
7	007-004	Shower Set	130x200x100mm	Antique Teak	30	30	2	10	1	1	8
8	007-004	Shower Set	130x200x100mm	Antique Teak	30	30	2	10	1	1	8
9	007-004	Shower Set	130x200x100mm	Antique Teak	30	30	2	10	1	1	8
10	007-004	Shower Set	130x200x100mm	Antique Teak	30	30	2	10	1	1	8
11	007-004	Shower Set	130x200x100mm	Antique Teak	30	30	2	10	1	1	8
12	007-004	Shower Set	130x200x100mm	Antique Teak	30	30	2	10	1	1	8
13	007-004	Shower Set	130x200x100mm	Antique Teak	30	30	2	10	1	1	8
14	007-004	Shower Set	130x200x100mm	Antique Teak	30	30	2	10	1	1	8
15	007-004	Shower Set	130x200x100mm	Antique Teak	30	30	2	10	1	1	8
16	007-004	Shower Set	130x200x100mm	Antique Teak	30	30	2	10	1	1	8
17	007-004	Shower Set	130x200x100mm	Antique Teak	30	30	2	10	1	1	8
18	007-004	Shower Set	130x200x100mm	Antique Teak	30	30	2	10	1	1	8
19	007-004	Shower Set	130x200x100mm	Antique Teak	30	30	2	10	1	1	8
20	007-004	Shower Set	130x200x100mm	Antique Teak	30	30	2	10	1	1	8
21	007-004	Shower Set	130x200x100mm	Antique Teak	30	30	2	10	1	1	8
22	007-004	Shower Set	130x200x100mm	Antique Teak	30	30	2	10	1	1	8
23	007-004	Shower Set	130x200x100mm	Antique Teak	30	30	2	10	1	1	8
24	007-004	Shower Set	130x200x100mm	Antique Teak	30	30	2	10	1	1	8
25	007-004	Shower Set	130x200x100mm	Antique Teak	30	30	2	10	1	1	8
26	007-004	Shower Set	130x200x100mm	Antique Teak	30	30	2	10	1	1	8
27	007-004	Shower Set	130x200x100mm	Antique Teak	30	30	2	10	1	1	8
28	007-004	Shower Set	130x200x100mm	Antique Teak	30	30	2	10	1	1	8
29	007-004	Shower Set	130x200x100mm	Antique Teak	30	30	2	10	1	1	8
30	007-004	Shower Set	130x200x100mm	Antique Teak	30	30	2	10	1	1	8
31	007-004	Shower Set	130x200x100mm	Antique Teak	30	30	2	10	1	1	8
32	007-004	Shower Set	130x200x100mm	Antique Teak	30	30	2	10	1	1	8
33	007-004	Shower Set	130x200x100mm	Antique Teak	30	30	2	10	1	1	8
34	007-004	Shower Set	130x200x100mm	Antique Teak	30	30	2	10	1	1	8
35	007-004	Shower Set	130x200x100mm	Antique Teak	30	30	2	10	1	1	8
36	007-004	Shower Set	130x200x100mm	Antique Teak	30	30	2	10	1	1	8
37	007-004	Shower Set	130x200x100mm	Antique Teak	30	30	2	10	1	1	8
38	007-004	Shower Set	130x200x100mm	Antique Teak	30	30	2	10	1	1	8
39	007-004	Shower Set	130x200x100mm	Antique Teak	30	30	2	10	1	1	8
40	007-004	Shower Set	130x200x100mm	Antique Teak	30	30	2	10	1	1	8
41	007-004	Shower Set	130x200x100mm	Antique Teak	30	30	2	10	1	1	8
42	007-004	Shower Set	130x200x100mm	Antique Teak	30	30	2	10	1	1	8
43	007-004	Shower Set	130x200x100mm	Antique Teak	30	30	2	10	1	1	8

Figure 9.10: Final Inspection Report

9.12. Reports and Their KTF

Table 6: Report and their KTF #

Document Name	KTF #
Allocation Slip	KTF 901
Cutting Issue Book	KTF 874
Daily Cutting Ledger	KTF 870
QC In Line Inspection Report	KTF 550
Daily Stitching Production Register	KTF 112
Daily Production Report	KTF 538
Pre Final Inspection Report	KTF 877
B Grade Fabric Report	KTF 875

Chapter 10: Lead Time Calculation

I have calculated the Lead Time in unit 2 of the stitching section in line 7 of CM 751 of the article pillow. In this activity my main aim was to specify the value added and non-value added times for which I have designed a complete Excel Sheet showing each and every detail of each operation involved in stitching a pillow. Here I tried to find the percentage of value added and non-value added time.



Lead Time Calculation

Dated: 20/07/2024

Details

Sr #	Type	Value
1	Section	Stitching
2	Unit	2
3	Line	7
4	CM	751
5	Article	Pillow
6	Size	50*75" 15 cm flap
7	Color	Red
8	Design	Christmas Berris
9	Customer	Laura Ashley



Detail Time Study

Sr #	Activity	Working	Moving	Waiting	Value Added/Non-Value Added	Start Time	End Time	Time Taken	Value Added Time	Non-Value Added Time
1	1 cm hem+ Label on Flap				Value Added	11:17 AM	11:17 AM	14.04		
2	Waiting				Non-Value Added	11:17 AM	11:37 AM	1169.1		
3	Moving				Non-Value Added	11:37 AM	11:37 AM	8.38		
4	Waiting				Non-Value Added	11:37 AM	11:55 AM	1046.71		
5	Front+Back+Flap Layering				Value Added	11:55 AM	11:56 AM	24.62		
6	Waiting				Non-Value Added	11:56 AM	3:22 PM	4536.74		
7	Moving				Non-Value Added	3:22 PM	3:23 PM	7.63		
8	Front+Back+Flap 4 side Stitch				Value Added	3:23 PM	3:26 PM	94.79		
9	Waiting				Non-Value Added	3:26 PM	3:27 PM	183.29		
10	Overlock 4 sides				Value Added	3:27 PM	3:27 PM	13.07		
11	Total(Seconds)	146.52	16.01	6935.84				7098.37	146.52	6951.85
12	Total(Minutes)	2.442	0.266833	115.5973				118.306167	2.442	115.8641667
13	Total(Hours)	0.0407	0.004447	1.926622				1.97176944	0.0407	1.931069444

*Break Time of 2 Hours is not included in this time study which is from 12 pm to 2 pm

% Value Added Time

2.064136

% Non-Value Added Time

97.935864

Total Lead Time

1.97176944

Figure 10.1: Lead Time Calculation

Here I specified each operations value added and non-value added time and ultimately found the total percentage of value-added and non-value added time.

Chapter 11: SAM

I have also calculated the total days required for stitching the duvy set and bedsheet set using SAM of different set being made in 8 hours and 12 hours work shift. Then I analyzed the difference of work shift effecting the total days required for stitching process.

Problem: Find out the Total days required to stitch the Duvy set and Bedsheet set.

1 Line 32 operators

Case 1:	8	Hours Shift
----------------	---	-------------

Article 1: Duvy Set having Nairobi Design				
	1 Duvy	2 Pillow	Total	
Quantity:	20000	40000		
SAM:	9.5	5.5		
Total SAM:	190000	220000		
Min per Day:	15360	min/day		
Days:	12.369792	14.322917	26.69271	days

Article 2: Bedsheet Set having Garden Green Design				
	1 Bedsheet	1 Pillow	1 Fitter	Total
Quantity:	15000	30000	15000	
SAM:	8.5	6	9	
Total SAM:	127500	180000	135000	
Min per Day:	15360	min/day		
Days:	8.3007813	11.71875	8.789063	28.80859375

Article 3: Duvy Set having Garden Blue Design				
	1 Duvy	2 Pillow	Total	
Quantity:	30000	60000		
SAM:	8.7	5.5		
Total SAM:	261000	330000		
Min per Day:	15360	min/day		
Days:	16.992188	21.484375	38.47656	days

Article 4: Bedsheet Set having Flower Pink Design				
	1 Bedsheet	1 Pillow	1 Fitter	Total
Quantity:	40000	80000	40000	
SAM:	8.5	6	9	
Total SAM:	340000	480000	360000	
Min per Day:	15360	min/day		
Days:	22.135417	31.25	23.4375	76.82291667

Total days to stitch Duvy Set:	65.16927083
Total days to stitch Bedsheet Set:	105.6315104
Total Days to stitch the 4 set of Articles:	170.8007813

1 Line 32 operators

Case2 :	12	Hours Shift
----------------	----	-------------

Article 1: Duvy Set having Nairobi Design			
	1 Duvy	2 Pillow	Total
Quantity:	20000	40000	
SAM:	9.5	5.5	
Total SAM:	190000	220000	
Min per Day:	23040	min/day	
Days:	8.2465278	9.5486111	17.79514 days

Article 2: Bedsheet Set having Garden Green Design			
	1 Bedsheet	1 Pillow	1 Fitter
Quantity:	15000	30000	15000
SAM:	8.5	6	9
Total SAM:	127500	180000	135000
Min per Day:	23040	min/day	
Days:	5.5338542	7.8125	5.859375
			19.20572917

Article 3: Duvy Set having Garden Blue Design			
	1 Duvy	2 Pillow	Total
Quantity:	30000	60000	
SAM:	8.7	5.5	
Total SAM:	261000	330000	
Min per Day:	23040	min/day	
Days:	11.328125	14.322917	25.65104 days

Article 4: Bedsheet Set having Flower Pink Design			
	1 Bedsheet	1 Pillow	1 Fitter
Quantity:	40000	80000	40000
SAM:	8.5	6	9
Total SAM:	340000	480000	360000
Min per Day:	23040	min/day	
Days:	14.756944	20.833333	15.625
			51.21527778

Total days to stitch Duvy Set:	43.44618056
Total days to stitch Bedsheet Set:	70.42100694
Total Days to stitch the 4 set of Articles:	113.8671875

Chapter 12: SWOT Analysis

12.1 Strengths

The greatest strength of KTML is that it produces maximum of its own products like yarn, fabric instead of buying it from other manufacturers. This is helping KTML to improve quality of their products day by day to increase customer satisfaction.

12.2 Weakness

The weakness is that the KTML is running on the paper documentation process, which is quite difficult to maintain record, and to maintain record in a decent and easy way it is important for KTML that they must shift to ERP systems for maintaining and securing the record of reports.

12.3 Opportunities

Opportunity lies in the field of supply chain, lean, innovative thinking and process improvement. Opportunities also lies in the quality sector to attract the customers.

12.4 Threats

Some other top Textile Companies like Interloop, Younus Textile and Nishat Mills are major competitors of KTML due to their Oracle based and smart ERP Systems.

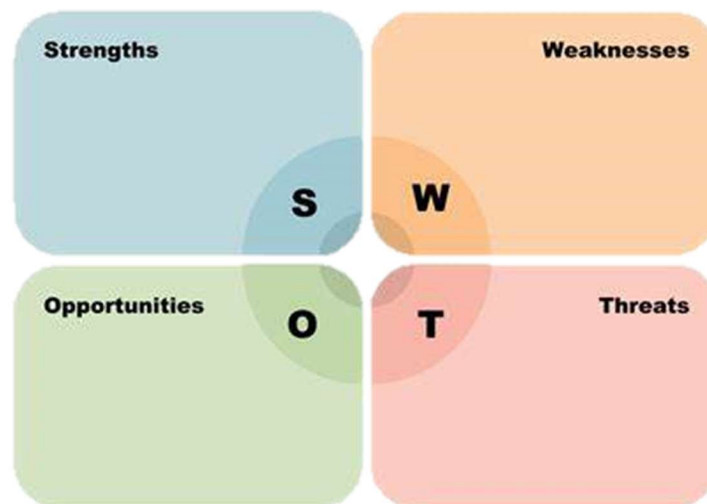


Figure 12.1: SWOT Analysis

Chapter 13: Recommendation

Converting the paper report into the oracle based or ERP based report system is important as it will help to keep record of report easy and as well as it is an easy method to maintain and enter the record. Developing and Implementing the ERP or Oracle based reports will help the company to safe old records as well as they can save the cost of paper and run on sustainability concept. Here is the recommendation for it:

13.1 Need Assessment

Identify specific reporting requirements and user needs.

13.2 System Selection

Choose an appropriate Oracle based or ERP solution.

13.3 Data Migration

Transfer existing paper based records to new digital system.

13.4 User Training

Provider training to user on new system.

13.5 Testing & Quality

Ensure the system meets requirement and functions correctly.

13.6 Go Live

Launch the new system and monitor progress.

13.7 Ongoing Support

Offer continuous support and maintenance to ensure system efficiency.